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Universal Cross-app Attacks: Exploiting and Securing OAuth 2.0 in Integration Platforms

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Integration Platforms such as Workflow Automation Platforms, Virtual Assistants and Smart Homes are becoming an integral part of the Internet. These platforms welcome third-parties to develop and distribute apps in their open marketplaces, and support "account linking" to connect end-users' app accounts to their platform account. This enables the platform to orchestrate a wide range of external services on behalf of the end-users. While OAuth is the de facto standard for account linking, the open nature of integration platforms poses new threats, as their OAuth architecture could be exploited by untrusted integrated apps.

In this paper, we examine the flawed designs of multi-app OAuth authorizations that support account linking in integration platforms. We unveil two new platform-wide attacks due to the lack of app differentiation: *Cross-app OAuth Account Takeover (COAT)* and *Request Forgery (CORF)*. As long as a victim end-user establishes account linking with a malicious app, or potentially with just a click on a crafted link, they risk unauthorized access or privacy leakage of any apps on the platform.

To facilitate systematic discovery of vulnerabilities, we develop COVScan, a semi-automated black-box testing tool that profiles varied OAuth designs to identify cross-app vulnerabilities in real-world platforms. Our measurement study reveals that among 18 popular consumer- or enterprise-

facing integration platforms, 11 are vulnerable to COAT and another 5 to CORF, including those built by Microsoft, Google and Amazon. The vulnerabilities render widespread impact, leading to unauthorized control over end-users' services and devices, covert logging of sensitive information, and compromising a major ecosystem in single click (a CVE with CVSS 9.6). We responsibly reported the vulnerabilities and collaborated with the affected vendors to deploy comprehensive solutions.

Category:

Short Presentation

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